



INTERNATIONAL UNMANNED AERIAL VEHICLES COMPETITION

TEAM BAYUSUTA

FREE MISSION CATEGORY
PROJECT PRESENTATION REPORT

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TEAM NAME: BAYUSUTA
TEAM TYPE: ☒ UNIVERSITY TEAM ☐ MIXED TEAM
VEHICLE TYPE: FIXED WING
VEHICLE DEVELOPMENT TYPE: ☒ NEW VEHICLE ☐ CURRENT VEHICLE
UNIVERSITY NAME: SEPULUH NOREMBER INSTITUTE OF TECHNOLOGY
COUNTRY: INDONESIA
TEAM RESPONSIBLE: RAMA SURYANSYAH BUDIANTO

1. SUMMARY

1.1 Team Organization

Bayusuta is an unmanned aerial vehicle (UAV) research team that originated from the Technology Development division of Bayucaraka ITS, focusing on research and development of systems and sub-systems to solve problem related in the field of UAV. Below in Figure 1 is the organizational diagram of the team, along with a detailed breakdown of each members roles and responsibilities in the development of the Free Mission aircraft.

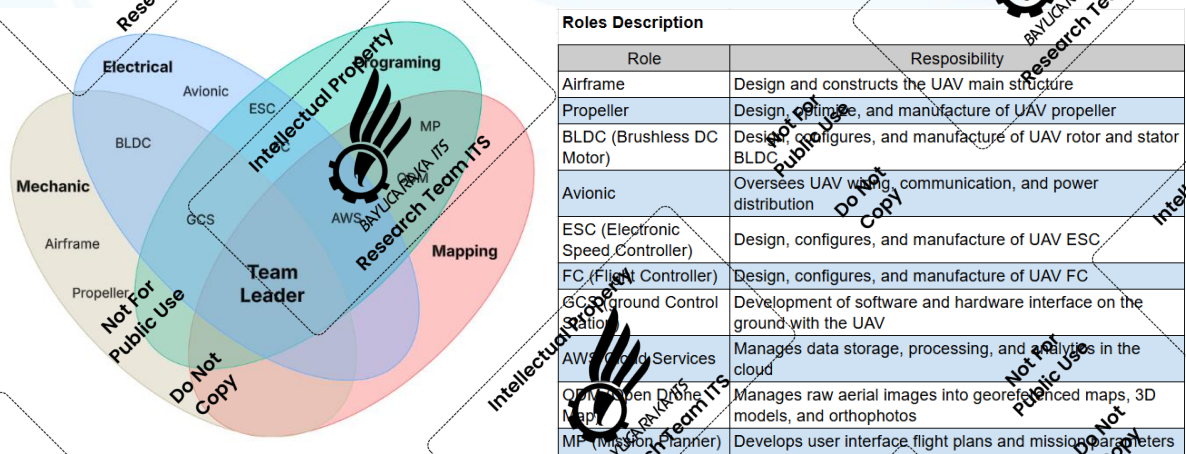


Figure 1. Team Bayusuta Organizational Diagram

The diagram illustrates the core areas of expertise and responsibility within Team Bayusuta comprise of Mechanics, Electrical, Programming, and Mapping. The overlapping sections represent collaborative efforts among the team members, with the Team Leader positioned at the center to coordinate and integrate all activities. The team itself is composed of 10 third-year university students, each bringing a diverse set of skills and experiences to effectively address the complex challenges of Bayusuta Free Mission aircraft.

1.2 Mission Title

1.2.1 Mission Background

In today's modern era, rapid technological advancements continue to open up unprecedented opportunities to tackle challenges and explore ideas once deemed impossible. One such innovative technology is the unmanned aerial vehicle or UAV. The presence of UAVs revolutionizing various sectors by simplifying tasks once required a full-size aircraft to be done and incurred hefty price. One of those tasks is strategic information gathering and executing search and rescue (SAR) operations in difficult-to-access disaster zones. However, leveraging UAV technology effectively is not always straightforward, especially for end-users who lack the necessary understanding, direct experience, or time for complex mission aeromodeling.

1.2.2 Mission Problem

Based on the mission background discussed, Bayusuta team mission is to gather information on potential recent fire zones in affected areas with easy to operate procedure by the end-user. In our home country Indonesia is a country that boasts some of the most diverse land biodiversity regions in the world. The risk of fires in hard to access remote areas poses a significant challenge, and rapid operational response is essential to prevent these fires from spreading. To address this critical issue, deployment of advanced UAV technology which capable of providing timely and accurate data even in difficult to access areas enable early detection of new fire hotspots and delivers vital real-time information to end-user.

1.2.3 Mission Expected Outcome

Bayusuta fixed-wing aircraft is developed to address the mission problem discussed delivering an aircraft with functionality and user-friendly operation. The aircraft will incorporate Bayusuta original in house designed and build airframe, flight controller (FC), brushless DC motor, Electronic speed controller (ESC), and propeller for optimized aircraft size and flight efficiency. All of these components and system will work in unison and integrate seamlessly into a centrally operated software and handheld ground control station (GCS), ensuring real-time feedback and complete control over the aircraft during the ground setup phase or executing a mission in the air. The mission will output strategic information in the form of live video feedback and area map or Orthophoto and digital elevation models (DEMs) of the fire zones while the aircraft is in flight operation.

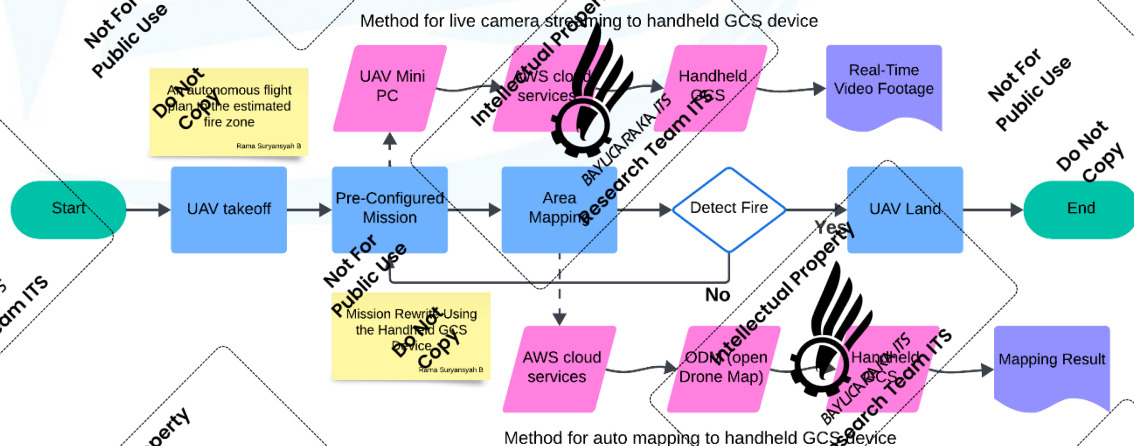


Figure 2. Mission Expected Outcome

2. MISSION INFORMATION

2.1 Flight Mission Description

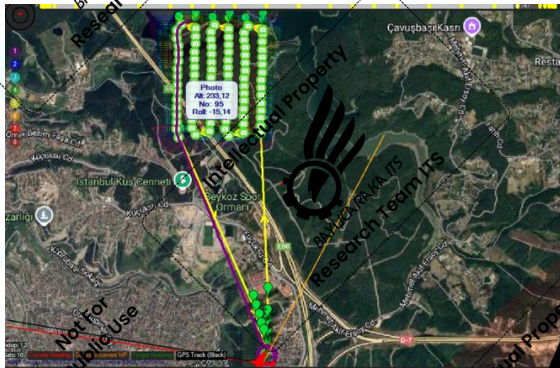


Figure 3. Flight Mission

Upon reaching the designated mapping area situated 2.8 km away from the launching zone a GoPro camera will then capture detailed imagery of the potential fire zones.

The collected imagery is then processed using OpenDroneMap (ODM) and shared with the handheld ground control station (GCS) for immediate analysis. This mission is designed with specific flight parameters in mind an operational altitude of 80 meters, and a cruise speed of 22 m/s. The designated mapping area covers 1 km x 1 km and the time to complete the mapping area is 14.5 minutes. Notably, the mission can be dynamically adapted by the user if no fire is initially detected within the pre-configured flight plan by uploading another pre-configured flight plan and executing it on the handheld device, allowing for flexible and responsive aerial surveillance with a maximum flight time of 40 minutes.

2.2 Flight Mission Originality

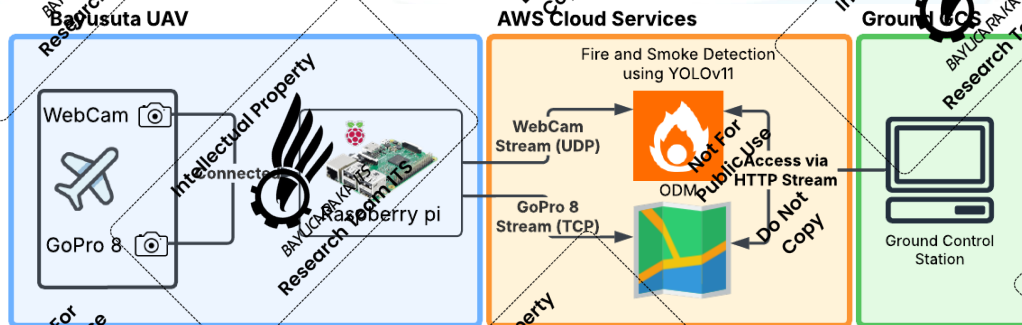


Figure 4. Mission Outline

The Bayusuta UAV mission primarily focuses on two main objectives: advanced fire and smoke detection, and auto-mapping. In order to achieve these objectives, the UAV utilizes two cameras, webcam and GoPro 8, each is assigned to do different tasks. The Webcam streams video transmitted from the Raspberry pi to AWS via User Datagram Protocol (UDP) for low-latency transmission, the computer vision model then processes the data for fire and smoke inference. The GoPro is utilized for capturing subsequent geotagged images, then transmitted via Transmission Control Protocol (TCP) to AWS to ensure data integrity. The images will be processed using OpenDroneMap (ODM) on AWS to generate accurate maps for analysis. Furthermore, the Ground Control Station (GCS) provides an interface for monitoring and mapping result retrieval from AWS, ensuring end-user interaction.

2.2.1 AWS Cloud Service

The Bayusuta UAV mission implements the use of AWS Cloud Services as the central computing unit for both real-time fire detection and auto-mapping. The system mainly uses AWS Elastic Compute (EC2) which serves as the primary processing unit for both objectives. By Leveraging AWS Cloud Services, the mission significantly reduces the need for high-cost onboard hardware, which are typically required for vision inference and mapping processes. Furthermore, utilizing the cloud service minimizes the risk of hardware failure and hardware limitations that might otherwise interfere with the UAV operations, ensuring reliable and efficient mission execution.

2.2.2 OpenDroneMap (ODM)

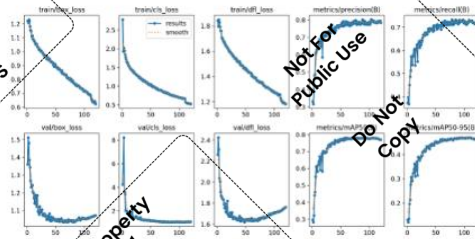


Figure 5. Orthophoto of Oyo River, Gunung Kidul, Yogyakarta, Indonesia

Once the images are uploaded the EC2 instance automatically initiates processing tasks using AWS Batch to orchestrate workload distribution to NodeODM a computing instance hosted on EC2 itself. NodeODM utilizes Docker containers to ensure consistency of the processing environment, avoid dependency conflicts, and accelerate algorithm execution. The core ODM algorithms, such as Structure from Motion (SfM) and multi-view stereo, then generate orthophotos as in Figure x with high resolution (4 cm/pixel) as well as digital elevation models (DEM/DSM) with 10 cm/pixel resolution. The final results are stored back in EC2 storage and can be processed via WebODM or integrated into handheld devices.

2.2.3 YOLOv11 Fire and Smoke detection

To detect fire and smoke in any environment, we utilized the YOLOv11 model and trained it on publicly available datasets. The model is fast and accurate, making it ideal for real-time detection tasks. For deployment, the model was converted using ONNX Runtime to ensure compatibility with hardware acceleration providers such as Intel OpenVINO, NVIDIA CUDA, AMD ROCm, and others. The inference process is performed using our software running on a cloud server GPU instance.



YoloV11 Small Version A

Class	Precision	Recall	mAP@50	mAP@50-95
all	0.789	0.789	0.789	0.469
Smoke	0.800	0.800	0.800	0.543
Fire	0.74	0.668	0.718	0.396

Figure 6. Performance Metrics of YOLOv11 Model

3. UAV INFORMATION

3.1 Design

3.1.1 Basic Configuration

Based on the mission, the UAV needs to achieve good endurance and good stability to carry out the mission. The Eppler-207 airfoil is used because of its performance in providing aerodynamic efficiency at low to moderate angles of attack.

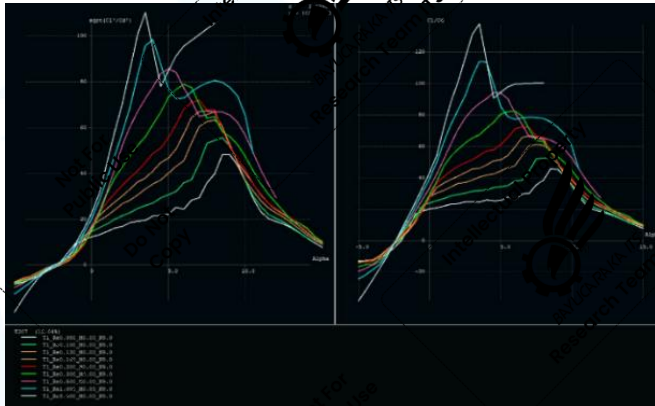


Figure 7. Eppler-207 Analysis

This image shows aerodynamic efficiency metrics for the Eppler 207 (E207) airfoil against the Angle of Attack (α) at various Reynolds numbers (Re). Figure 7. The left graph highlights endurance efficiency, peaking at 4-6° α . While the second graph (Lift-to-Drag Ratio, L/D) peaks at 3-5° α , optimizing cruise performance. On the other hand, it also shows that the Reynolds number applied to the airfoil is proportional to its aerodynamic efficiency and endurance efficiency.

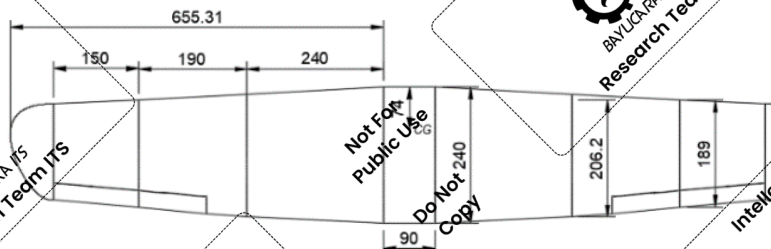


Figure 8. Wing Configuration

The image presents a detailed wing configuration that will utilize a moderate aspect ratio of up to 6.22 with a wingspan of 1400 mm, featuring a low-wing design. This design enhances stability by lowering the center of gravity while optimizing aerodynamic performance. Additionally, the positioning of the center of gravity (CG) at 74 mm above the reference line ensures balanced flight characteristics, while the structured division of the wing enhances rigidity and control effectiveness. On the other hand, V-tail configurations will be included to increase yaw stability and reduce aerodynamic drag, improving overall efficiency and control responsiveness.

Based on the wing configuration, the calculation of UAV's flight profile is obtained. The minimum stall speed (V_s) achieved is about 10.2 m/s, the maneuver speed (V_a) up to 24.08 m/s, 24.8 m/s of the cruise speed (V_c), and 33.8 m/s for the dive speed (V_d). The maximum theoretical bank load reaches 60°, with the turning radius as small as 17 m.

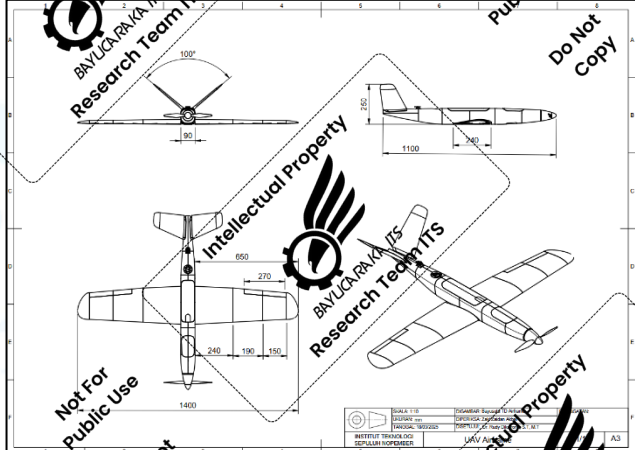


Figure 9. Technical Drawing

The consideration of each configuration for the UAV highly affects the performance and capability of the UAV. Based on the configuration that has been adjusted with the mission scope, the Bayusuta UAV is proposed as a fixed-wing aircraft with puller types of propulsion. The design is built using Fusion 360 and ANSYS to find out the capability of the overall design in a simulated environment. The mechanical design can be seen in the figure 9.

The component's placement can be seen in the figure 10. The location of each component is adjusted and distributed across the fuselage to make sure the Center of Gravity (CG) does not shift, thus maintaining its performance and comfort for the user. The respective figure also shows that there is a huge amount of space for the battery placement. This section is intended as an area to make a precise adjustment, thus keeping the center of gravity in its desired place.

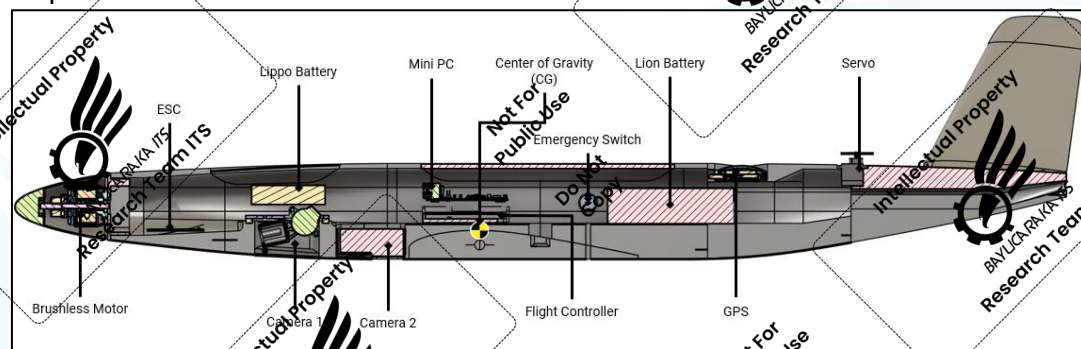


Figure 10. Components Placement

There are two cameras as a payloads, the first one is for photogrammetry which is permanently placed slightly in front of the CG. While the other one is placed next to the photogrammetry camera. Utilizing a stepper motor, this camera can be deployed using an axis gimbal to operate the mission as intended. The mechanism of this camera can be seen in the figure 11.

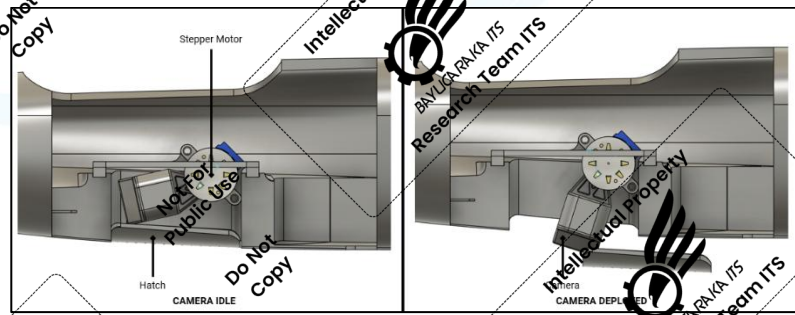


Figure 11. Payload Arrangement

3.2 Hardware

Bayusuta UAV system architecture is summarized in the following wiring diagram.

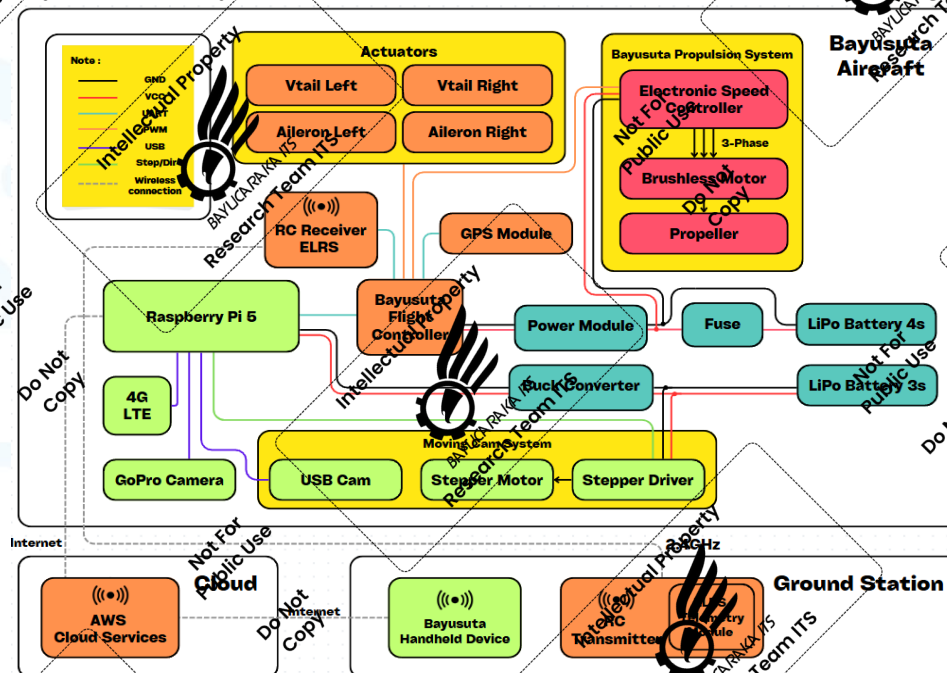


Figure 12. Wiring Diagram Bayusuta Aircraft

Where components are a combination of commercially available components such as a mini PC, GPS module, and RC receiver. It also features original Bayusuta developed components including flight controller (FC), brushless DC motor, electronic speed controller (ESC), propeller, and handheld ground control station (GCS).

3.3 Software

3.3.1 Ground Control Software (GCS)

The ground control software of the UAV is a proprietary system, consisting of both a frontend and a backend server. It is specifically designed to support the operation, monitoring, communication, and management of the UAV. The software integrates various data sources, such as GPS, flight position, and other vehicle state information, which are displayed in real-time on the ground station computer via the connected telemetry. In addition to these data sources, live video captured by the camera is transmitted in real-time to the AWS server, where it is processed and then sent back to our software, which will be displayed on the OSD (On-Screen Display).

3.3.2 WebODM

WebODM is an open-source web interface for the OpenDroneMap (ODM) system. It simplifies the process of transforming aerial imagery into valuable geospatial products such as orthophotos and digital elevation models (DEMs). Through a distributed computing capabilities, and Docker containerization WebODM provides a streamlined platform to manage, process, and visualize camera captured data. On the Bayusuta UAV system, all data management and processing is automated using algorithms written and executed within the AWS cloud services infrastructure.

3.4 Originality-Team Production

3.4.1 Airframe

The analysis of the wing and tail was conducted using XFLR5 as an aerodynamic analysis program. The E207 wing profile with a wingspan up to 1400 mm including the tail with NACA0010 is drawn as seen in the figure 13. The V-Tail was arranged with 100° deflection to ensure the UAV gains pitch and yaw control.

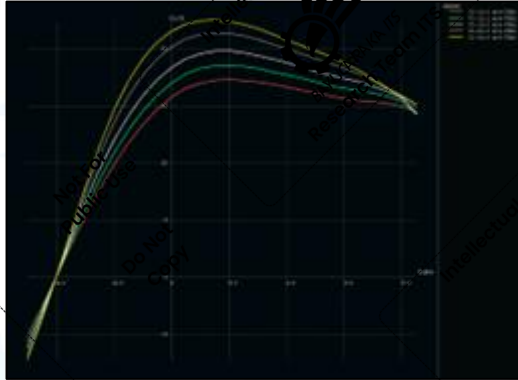


Figure 13. Aerodynamic Efficiency

The minimum angle of attack was set to 5° with a maximum of 15° with several velocities according to the flight envelope. The Cl/Cd ratio in each angle of attack was obtained which can be seen in figure 13. The graph shows that the aerodynamic efficiency of the aircraft peaks at α around 2° to 4° , which is optimal for cruise angle. On the other hand, the Cm vs α represents that the wing naturally nose down as the angle of attack increases, which is naturally stable.

Schrenk theory is used to analyze the lift distribution of the UAV, combining the elliptical and trapezoidal lift equations. Thus increase the approximation of the generated data. Maneuver speed (V_a) was used, which is 24.08 m/s. Figure 14 shows that the mean distribution in the root chord reaches up to 151.25 N/m and slowly decreases across the half-wingspan to 24.76 N/m. These value shows that the structure of the wing still withstand with the ongoing forces through the wingspan manufactured with PLA+ (Polylactid Acid) and carbon tube, whereas the maximum Von-Mises stress is 52.5 Mpa. The visualization of the lift distribution can be seen in figure 14.

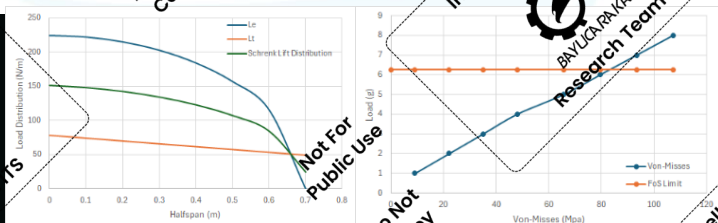
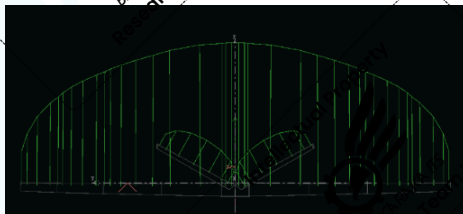
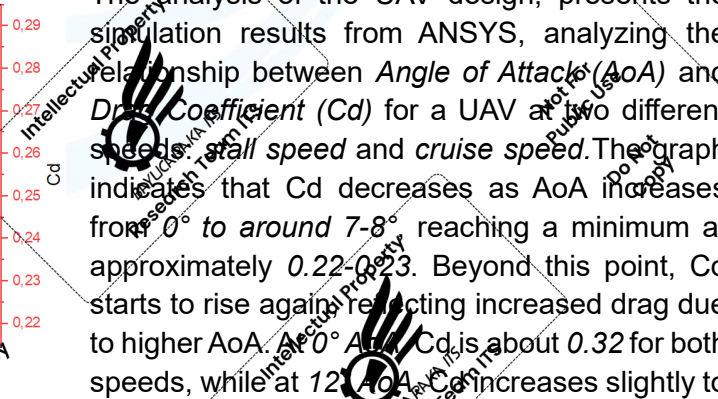
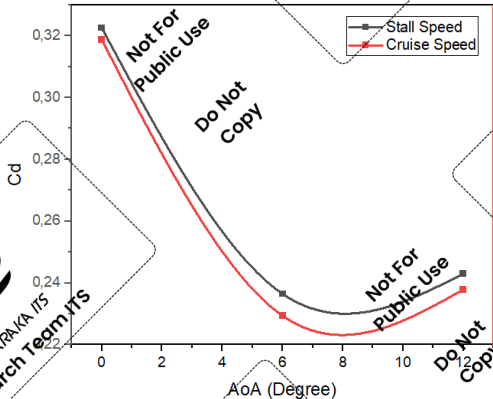


Figure 14. (Left) Lift Distribution; (Middle) Schrenk Lift Distribution; (Right) Von-Mises Stress

The analysis of the UAV design, presents the simulation results from ANSYS, analyzing the relationship between *Angle of Attack (AoA)* and *Drag Coefficient (Cd)* for a UAV at two different speeds: *stall speed* and *cruise speed*. The graph indicates that Cd decreases as AoA increases from 0° to around $7-8^\circ$, reaching a minimum at approximately 0.22-0.23. Beyond this point, Cd starts to rise again, reflecting increased drag due to higher AoA . At 0° AoA , Cd is about 0.32 for both speeds, while at 12° AoA , Cd increases slightly to around 0.23-0.24. The cruise speed condition consistently shows slightly lower drag than stall speed, indicating better aerodynamic efficiency. The optimal AoA for minimal drag is found near $7-8^\circ$, which is crucial for maximizing UAV performance.



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3.4.2.1 BLDC Motor

In designing a high-quality BLDC, it is necessary to pay attention and calculate to minimize BLDC powerflow losses. The winding, material and grade of magnetic play a crucial role in the quality of magnetic flux generation as well as the durability of the motor to perform high-RPM with minimum temperature.

3.4.2.1 Motor Specifications and Parameters

Stator core in this BLDC motor is made from non-grained electrical steel material grade steel NV30S-120 with a thickness of 0.35mm. This electrical steel material was chosen because it has high permeability, has low core loss, and can withstand higher temperatures without reducing its magnetic properties. The physical and mechanical characteristics of the material used are as follows.

Table 1. Physical Characteristics of Core Materials

Specification	Value	Unit
Material Type	Electrical Steel	-
Grade	NV30S-120	-
Thickness	0,35	mm
Density	7750	Kg/m ³
Bulk Conductivity	3333330	Siemens / m
Specific Heat	460	J/kg °C
Young's Modulus	2,1 x 10 ¹¹	N/m ²
Poisson's Ratio	0,3	-

A combination with a low number of slots and poles is used to achieve high speeds. The determination of the number of slots and poles also considers the dimensions of the motor and the ease of prototyping. A combination of 12 slots and 14 poles with a winding factor of 0.933.

3.4.2.2 Stator Winding Configuration

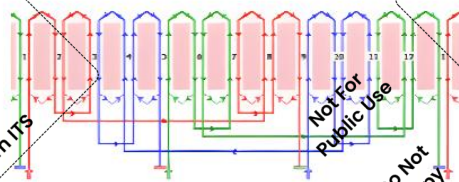
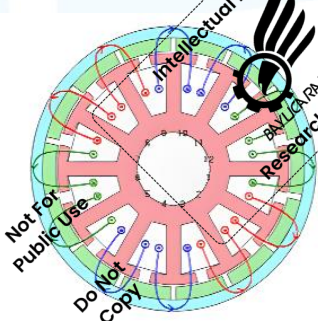


Figure 15. Stator Winding

The coil consists of a 3-phase coil connected in delta. The delta connection is chosen to obtain a low voltage source. The motor has 14 pairs of poles, length 20mm, radius 21.5mm, field strength 0.4 T. Determined the number of turns is 6 turns. The copper diameter chosen is 0.2mm (AWG 32) with a current capacity of 0.53 A. AWG 32 copper has a maximum frequency to achieve 100% skin depth of 430kHz, so the skin effect can be ignored. combination of 12 Slots and 14 Poles, the calculation of q factor is as follows.

$$q = \frac{N_p}{\text{ph. N}} = \frac{12}{3 \times 14} = 0.285$$

From the q factor results obtained, a concentrated winding type is used.

3.4.2.2 Simulation Results of BLDC Motor in Ansys Maxwell

After a simulation of the BLDC motor using Ansys Maxwell 2024 software, the results are obtained in the form of phase current and induced voltage in the BLDC motor.

3.4.2.3.1 Phase Current

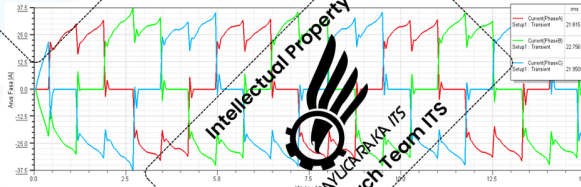


Figure 16. Graph of Phase Current and Time

Next, a graph of phase current against time is obtained. Based on Figure 16, the rms value of the phase current of the BLDC motor is about 21.81A, 22.75A, 21.95A in each phase at a simulation time of 15ms. Each phase at a simulation time of 15ms. The phase current values are almost balanced and have a difference below 1 A.

3.4.2.3.2 Induced Voltage

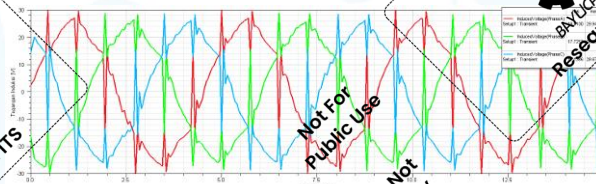


Figure 17. Graph of Induced Voltage and Time

Induced voltage or known as Back EMF is one of the most important parameters in supporting motor performance parameter that is very important in supporting motor performance. Based on the simulation results, the induced voltage characteristics are outlined in graph form. Induced voltage graph against time can be seen in Figure 17.

3.4.2.4 BLDC Motor Design

After determining the parameters, continue modeling the desired BLDC motor with a 3D design to project and as a guideline for the manufacturing process. with 3D design to project and guide the manufacturing process. 3D design motor using Fusion 360.

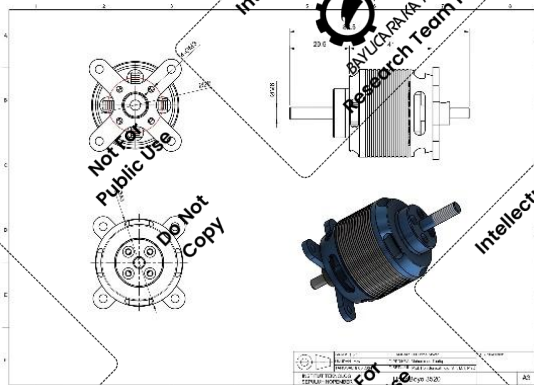


Figure 18. BLDC Technical Drawing

The airflow design on the motor can circulate 2.5x more optimally to create a self-cooling system during operation and because of that the motor has dust-resistance. BLDC motor is designed with high-temperature enameled wire, which has a resistance of up to 220°C. The motor can operate at high-RPM without worrying about the heat on the motor. In addition, the windings are insulated with a special insulating layer so that the motor has a waterproof advantage.

over a space for anodizing the motor base and zinc plating for the rotor case. This zinc plating treatment on the rotor will also provide the benefit of increasing the rotor's resistance from corrosion for 35 years, as well as increasing the thermal quality up to 1000°C.

3.4.3 Electronic Speed Controller (ESC)

Designing an Electronic Speed Controller (ESC) to support the mission objectives requires a propulsion system with high efficiency and reliability. In the context of ESC, this can be achieved through measurement, thorough analysis, and optimization of the circuit or firmware. Some of these analyses include, but are not limited to.

3.4.3.1 Power Losses Analysis

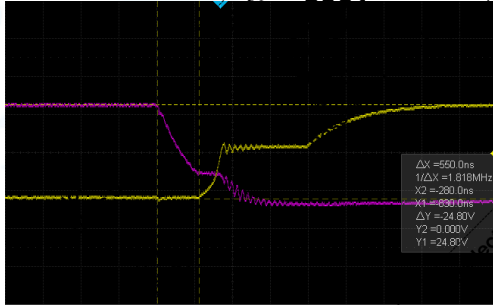


Figure 19. High-phase and Low-phase switching Measurement

Ideally power losses in a half-bridge circuit should be zero, but in practice this is impossible. That's why it is crucial to design an ESC to minimize these losses. Key strategies that can be done reducing MOSFET losses such as switching, conduction, body diode, and gate charge losses. Implementing an ideal dead time switching between high-phase and low-phase as shown in Figure 19. avoiding shoot-through help minimizing unnecessary power losses and dissipation.

3.4.3.2 Power Dissipation Analysis

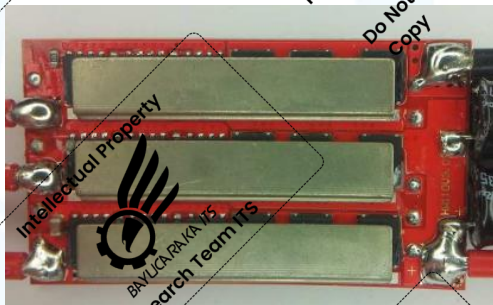


Figure 20. Application of Busbar

limitations in PCB dimensions can hinder MOSFET regulation especially under high current loads. To address this without increasing PCB size, key strategies that can be done placing multiple parallel vias across the whole 4-layer PCB with an inner ground plane and installing busbars to the MOSFET source terminal as shown in Figure 20. effectively increasing PCB area. these approaches can enhance both current and thermal dissipation.

3.4.3.3 Design Consideration

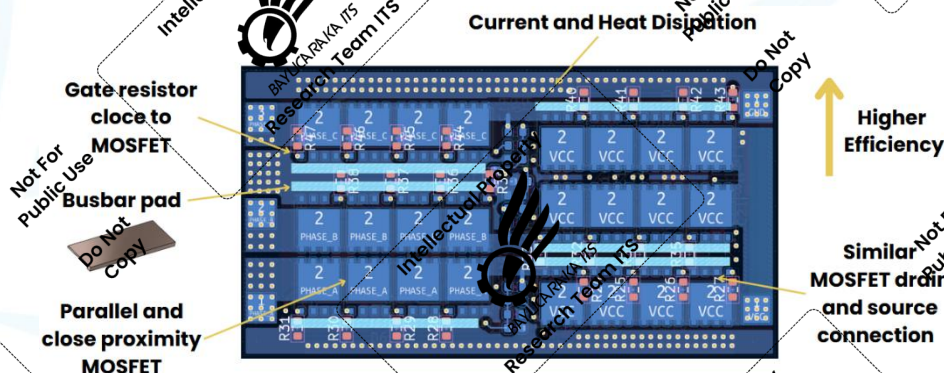


Figure 21. ESC Design Consideration

With all the aspects analyzed a high-efficiency ESC can be designed with a dimension of 70x38x1.2 mm. This ESC is specifically designed for the Bayusuta propulsion system, operating with a 4S battery and capable of reaching burst current up to 60A with minimal heat.

3.4.4 Propeller

Folding Propeller is a type of propeller that features a folding mechanism at the hub. This propeller utilizes centrifugal force, allowing the propeller blades to open when spinning and close when not in motion. This type of propeller offers several advantages such as being lighter in weight, more efficient, and reducing the risk of propeller damage during landing. Bayusuta folding propeller design is a modified 12x6 propeller made of PLA+. This modification involves adjusting the thickness of the propeller to match the material used, ensuring the desired durability is achieved.

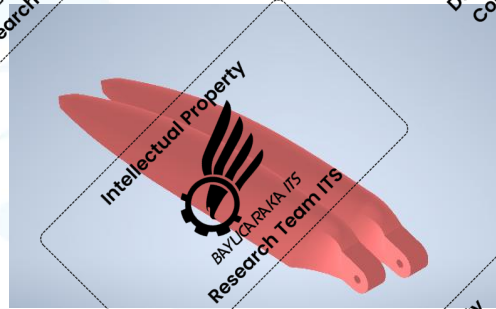


Figure 22. Folding Propeller Design

In the performance analysis of this propeller a comparison is made between the 12x6 Bayusta folding propeller and the APC 12x6 propeller based on material weight, thrust, and efficiency.

Table 1. Characteristics of the 12x6 Propeller

Propeller 12x6		
Parameter	Modified Folding Propeller	APC Propeller
Material	PLA+(Esun)	Nylon Plastic
Density	1,23 gr/cm ³	1,13 gr/cm ³
Weighth	26 gr	1.62 oz / 45,92623 gr
Tensile Strength	60 Mpa	80 Mpa
Flexural Strength	85,5 Mpa	112Mpa

Table 2. Comparison bench Test Data of the 12x6 Propeller

APC PROPELLER 12x6			Modified Folding Propeller 12x6		
Trottle (%)	Ampere (a)	thrust (gram)	Trottle (%)	Ampere (a)	Thrust (gram)
40	17,2	1117	40	12,2	1148
50	18,68	1406	50	16	1328
60	19.84	1670	60	20,5	1660
70	24.53	2155	70	25,6	1950
80	33.36	2310	80	34,06	2280
90	42.72	2810	90	44,45	2700
100	47.28	2992	100	56,04	3192

From the test bench data it was found that the peak modified folding propeller achieved a peak thrust of 3,198 grams, 55.67 amperes, and 773.9 watts of power. In comparison, the APC 12x6 propeller had a peak thrust of 2,992 grams, 47.28 amperes, and 685.84 watts. The higher thrust of the modified folding propeller is due to the optimization of the airfoil thickness ratio making it more efficient. Additionally, the folding propeller is 56.6% lighter, reducing the load on the motor and allowing it to reach higher RPMs. Based on this data, the folding propeller has an efficiency of 4.12 grams/watt, while the APC propeller has an efficiency of 4.36 grams/watt, which is quite similar despite the higher thrust performance.

3.4.5 Flight Controller

Figure 23. shows the PCB design of the Flight Controller which was designed using KiCad Software. This Flight controller have PCB size (P x L x T) : 56 x 85 x 1.8 mm. In figure 23. we can view the components contained in flight controller, namely STM32F427VGT, UWB DW1000 module, SD card slot, sensor BMI088, HMC5883L, BMP388. The sensor contained in this Flight Controller consists of a 3-axis accelerometer, 3-axis gyroscope, 3-axis magnetometer and barometer. Our objective was to create a low cost, high-performance flight controller. We are using STM32F427VGT for its performance and cost, integrated with UWB DW1000 module for indoor positioning so this Flight Controller can be used for another research project, SD card slot for data logging, 11 PWM channel for servo control, 3 UART and 1 I2C port for serial communication.

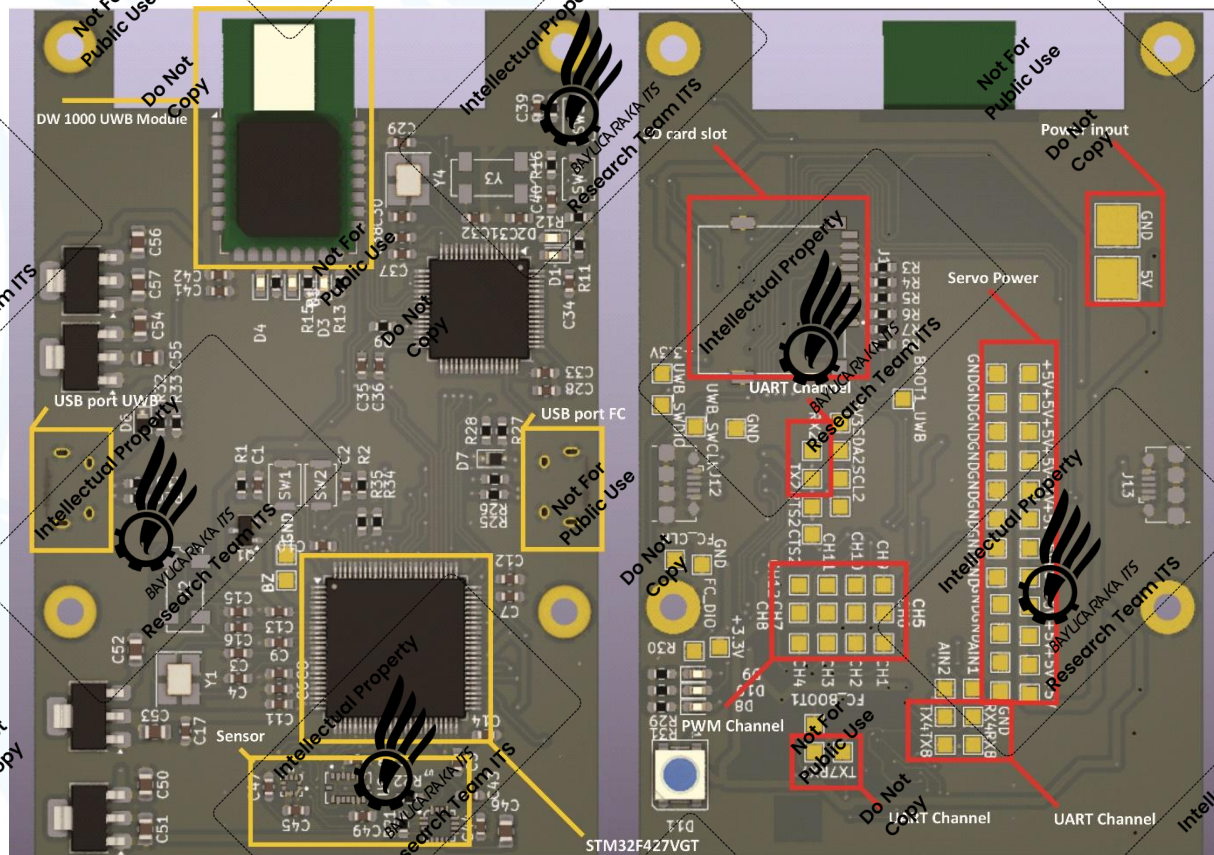


Figure 23. Flight controller bottom and top view

3.4.6.1 Ground Control Software

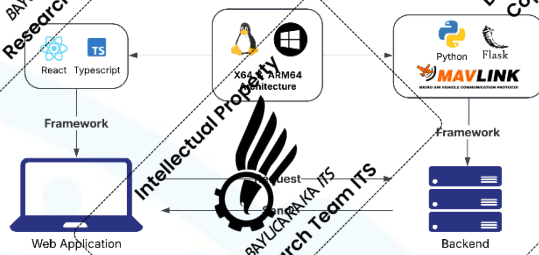


Figure 24. System Architecture



Figure 25. User Interface of the Application

3.4.6.2 Ground Control Handheld Device

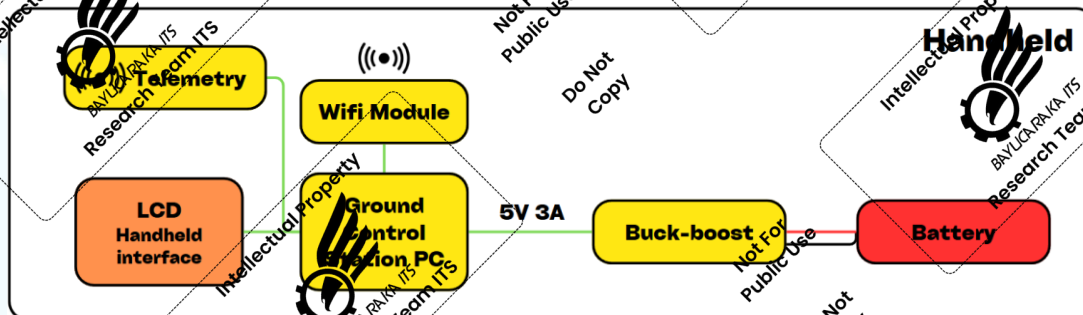


Figure 26. Handheld Ground Control Station (GCS) Wiring

The handheld Ground Control Station (GCS) is a portable, one-handed device designed for efficient and practical UAV mission control, providing precise and responsive aircraft operation through an intuitive interface. The device features a mini PC powered by an Intel N100 processor and a 10.1" capacitive touchscreen for seamless interaction. It also includes a telemetry port and a built-in Wi-Fi module, allowing users to optionally connect a telemetry device or enable Wi-Fi for UAV connectivity. Through this connection, the device accesses real-time data from the UAV and processed data from the aircraft's mini PC on AWS cloud services, allowing users to make informed decisions.

The device case is 3D-printed using PLA+ material, ensuring a lightweight, durable, and precisely crafted design. This method allows for efficient production and design flexibility while providing optimal protection for internal components. Additionally, the case includes a mechanical restart button directly connected to the mini PC for quick system reboots when needed.

4. COST ANALYSIS AND BUDGET TABLE

Materials and services that will help Team Bayusuta research and develop the UAV.

Table 4. Bayusuta Financial Support

No	Material/Services Planned to be Purchased	Category	Piece	Unit Price (IDR)	Unit Price (TRY)	Total Amount (TRY)	Reason for Purchasing Material / Services
1	PLA+ Filament	Mechanical	10	Rp216,000	€479.54	€4,795.20	for the manufacture of 4 aircraft, propeller research & manufacture, and other necessary components like in handheld devices
2	TPU Fiament	Mechanical	3	Rp228,000	€506.16	€1,518.48	for the linkage of control surfaces on the aircraft and other necessary components such as the FC casing
3	EMAX E08MA II Mini Sero	Flight Essential	16	Rp93,500	€96.57	€1,545.12	as the control surfaces mover of 4 aircraft
4	Pushrod stainless steel	Flight Essential	16	Rp44,000	€4.44	€71.04	as the control surfaces mover of 4 aircraft
5	carbon tube 3x10mm 11m	Flight Essential	2	Rp299,700	€299.70	€599.40	for the central structure of the aircraft wing
6	Raspberry Pi 5	Programming	1	Rp4,440,000	€4,440.00	€4,440.00	as the main FC unit in the aircraft
7	Micro stepper motor	Mechanical	2	Rp82,14	€82.14	€164.28	as the actuator of the tilting camera system
8	DRV8825 stepper driver	Mechanical	2	Rp84.36	€84.36	€168.72	as a driver for the tilting camera system
9	Webcam Logitech brio 500	Programming	1	Rp4,218,000	€4,218.00	€4,218.00	as the main camera for live video monitoring and fire detection
10	GoPro Hero 8	Mapping	1	Rp8,325,000	€8,325.00	€8,325.00	as the main camera for mapping
11	Liion 4s 8000mAh battery	Flight Essential	2	Rp1,665,000	€1,665.00	€3,330.00	The main avionics system battery is prepared as 2 spares
12	Liion 4s 2100mAh battery	Flight Essential	2	Rp359,64	€359.64	€719.28	the main battery of the mini PC system is prepared with 2 spares
13	Handheld Touch Screen	GCS	1	Rp4,218,000	€4,218.00	€4,218.00	Upgrade handheld device screen to capacitive touch providing cloud service user for data transmission, image processing, running computer vision programs, etc. according to mission requirements
14	Amazon Services	Programming	1	Rp4,440,000	€4,440.00	€4,440.00	as the receiver and telemetry used in the aircraft
15	Radiomaster Rp3 ExpressELRS	Flight Essential	1	Rp732.60	€732.60	€732.60	as the GPS used in the aircraft
16	Holybro M10 GPS	Flight Essential	1	Rp2,504.16	€2,504.16	€2,504.16	essential flying needs such as cables, connectors, buck converters, etc. according to the electrical system requirements
17	Electrical flight essential	Electrical	1	Rp1,110.00	€1,110.00	€1,110.00	essential manufacturing needs such as glue, adhesive, chasers, etc. according to manufacturing needs
18	Mechanical manufacture essential	Mechanical	1	Rp1,110.00	€1,110.00	€1,110.00	manufacturing services for rotor casing, cap and base for already own materials
19	BLDC manufacture services	BLDC	1	Rp4,662.00	€4,662.00	€4,662.00	all component used from LCSC, PCB printing services and assembly from JLCPCB for making 5 ESC
20	ESC manufacture and assembly	ESC	1	Rp7,104.00	€7,104.00	€7,104.00	all component used from LCSC, PCB printing services and assembly from JLCPCB for making 5 FC
21	FC manufacture and assembly	FC	1	Rp9,213.00	€9,213.00	€9,213.00	internet device that will be used on the aircraft for interconnection to AWS
22	D-Link DWR-X210	Flight Essential	1	Rp12,210.00	€12,210.00	€12,210.00	
TOTAL PROJECT SUPPORT REQUESTED FROM TUBITAK						€77,113.92	

